Marco Schouten

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Education

KTH Royal Institute of Technology — Stockholm, Sweden
Master of Science in Machine Learning, Grade 4.84/5.0 — 08/2020 - 10/2022

Delft University of Technology — Delft, Netherlands
Exchange (Masters) in Computer Science — 08/2021 - 01/2022

University of Genoa — Genoa, Italy
Bachelor of Science in Computer Engineering, Grade 109/110 — 09/2017 - 07/2020

Experience

Aptiv PLC — Wuppertal, Germany
Machine Learning Researcher — 03/2023 - Present

- Collaborating with a team of engineers to develop a customized neural network architecture that improves accuracy of high-definition (HD) maps.

SMaRC Swedish Maritime Robotic Center — Stockholm, Sweden
Research Intern — 01/2022 - 09/2022

- Developed and implemented a novel algorithm to improve the accuracy of sonar-based pose estimation for autonomous underwater vehicles, reducing drift and achieving accuracy up to 1m.
- Performed detailed analyses on the impact of various parameters such as sensor noise, and navigation drift providing valuable insights into optimizing system performance.
- Conducted extensive experiments and simulations on multiple underwater environments to validate the effectiveness of the proposed approach.

SurgiQ Ltd — Genoa, Italy
Research Intern — 03/2020 - 07/2020

- Developed and implemented an intelligent scheduling system for a hospital, resulting in 97% efficiency.
- Conducted thorough research on inductive AI techniques to inform the development of the scheduling system, leading to its publication on AI*IA and CLIC.

Skills

Programming	Python, Docker, Google Cloud Platform (GCP), Git
Deep Learning Frameworks	PyTorch, JAX, TensorFlow, Keras
Data Manipulation and Analysis	Pandas, MongoDB, SQL
Languages	English, Italian, Dutch (basic)

Awards

2022	**Selected at Nova**, a top 3% merit-based talent network for career development	Sweden
2020	**Hattrick Award in Artificial Intelligence**, awarded to 20% of the students [Link Award]	Sweden
2020	**Selected at Lead the Future**, mentorship with a 20% acceptance rate nationwide	Italy
2020	**Scholarship at University of Genoa 2018, 2019,2020**, ranked 4th out of 150+ students	Italy

Publications

Cademartori, L., Galatà, G., Monaco, C.L., Maratea, M., Mochi, M. and Schouten, M., **2022. An ASP-based Approach to Master Surgical Scheduling.** CILC 2022: 37th Italian Conference on Computational Logic

Alviano, M., Bertolucci, R., Cardellini, M., Dodaro, C., Galatà, G., Khan, M., Maratea, M., Mochi, M., Morozan, V., Porro, I. and Schouten, M., **2020. Answer Set Programming in Healthcare: Extended Overview.** IPS-RCRA@ AI*IA.

Projects

Real Time Object Detection | KTH Stockholm, Sweden
Python 05/2022 - 06/2022

- Managed intensive experiments by training deep neural networks utilizing Python, YOLOv5, reaching 45 FPS and 61% mAP for real-time object detection.
- Optimized DNN model to identify 4 different classes on 10K labeled image.
  [github.com]

Imitation learning for human-robot skill transfer | TU Delft Delft, Netherlands
Python 09/2021 - 12/2021

- Engineered Machine Learning-based software that accurately transferred hand movements captured by kinect to a simulated manipulator, raising accuracy rate to 91%.
- Designed creative and dynamic training module outlined with metrics and quantitative data analysis.

Training time estimation for CNN in Cloud Distributed Systems | TU Delft Delft, Netherlands
Python, Docker 08/2021 - 11/2021

- Utilized predictive modeling to optimize CNN training jobs, completing 33% more within pre-defined timeline utilizing 4 nodes and 16 GB of RAM.
  [github.com]

Starcraft AI Bot Competition | KTH Stockholm, Sweden
C++ 04/2021 - 06/2021

- Developed behavior tree algorithm in C++ & BWAPI for controlling multi-agent Starcraft AI, resulting in 50% win rate with 50+ simulation matches.
  [github.com]